Chapter

Waste Management

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1. Which of the following is a biodegradable waste? MAIN 2024

(a) Broken glass
(b) Wastepaper
(c) Polythene
(d) Plastic bags

2. Which of the following method of waste disposal is harmful? MAIN 2023

(a) Composting
(b) Segregation
(c) Dumping
(d) Vermicomposting

3. Explain how waste accumulation can be harmful. (Any two points) MAIN 2024

4. Give two uses of composting. MAIN 2024

5. Why are landfills set up far away from the city? (Any two reasons) MAIN 2024

6. Why should Waste Management be practiced in every school? MAIN 2023

7. Mention one advantage and one disadvantage of dumping of waste. MAIN 2023

8. How is recycling of waste helpful? Give an example of recycling of waste. MAIN 2023

9. Mention any one initiative taken by the Government to manage waste MAIN 2023

10. How can waste be reused ? Explain with the help of an example.
COMP 2018,2022

11. Name two toxic particulate materials. State the effect of each on human health. MAIN 2023

12. Why should we avoid using polythene carry bags and styrofoam cups ?
SQP 2021

13. What service is indirectly done by the ragpickers for the disposal of waste?
COMP 2022

14. Explain the role of segregation of waste in the safe disposal of waste.
MAIN 2023

15. Briefly answer the following: MAIN 2023
(i) How is segregation of the waste helpful?
(ii) Mention one way in which we can reuse waste.
(iii)Mention one benefit of Composting

16. Why is there a need for waste management in large metropolitan cities.
SQP 2022

17. Mention some benefits of composting. MAIN 2022

18. _____ is a way of disposing waste on land without causing health hazards or public safety. COMP 2021

(a) Dumping
(b) Sanitary landfill
(c) Recycling
(d) Incineration

19. ______ is an aerobic method of decomposing solid wastes. MAIN 2022

(a) Incineration
(b) Recycling
(c) Compositing
(d) Dumping

20. The organic waste from households undergo _____ to form compost.
SQP 2023

(a) Reduction
(b) Burning
(c) Decomposition
(d) Churning

21. Compositing rejuvenates poor soils by adding COMP 2021

(a) Nutrients
(b) Manure
(c) Humus
(d) Oxygen

23. Give a reason for the following : COMP 2022

(i) plant more trees.
(ii) use organic manure instead of chemical fertilizers.
(iii)use mass transport system.

24. Name the processes involved in reducing the waste. MAIN 2002

25. What is composting ? Give two advantages of using compost. SQP 2021

Solution

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1. (b) Wastepaper

Wastepaper is biodegradable because it can be decomposed naturally by microorganisms, unlike glass, polythene, or plastic bags.

2. (c) Dumping

Dumping waste without proper treatment is harmful to the environment as it leads to pollution of soil, water, and air. Other methods like composting, segregation, and vermicomposting are eco-friendly.

3. 1. Environmental Pollution: Accumulated waste can pollute air, water, and soil by releasing harmful chemicals and toxins, affecting human health and biodiversity.

4. Health Hazards: Untreated waste attracts pests like rodents and insects, spreading diseases and creating unhygienic living conditions.

5. Give two uses of composting.

1. **Soil enrichment:** Compost improves soil structure and provides essential nutrients, promoting healthy plant growth.
2. **Waste reduction:** Composting decreases the amount of organic waste sent to landfills, helping in effective waste management.

6. Why are landfills set up far away from the city? (Any two reasons)

1. To prevent **health hazards** and reduce the risk of diseases caused by waste.
2. To avoid **pollution** of air, water, and soil in populated areas, keeping the city environment clean.

7. Why should Waste Management be practiced in every school?

1. **Environmental responsibility:** It teaches students to reduce waste, recycle, and protect the environment, promoting eco-friendly habits from a young age.
2. **Clean and healthy surroundings:** Proper waste disposal keeps the school environment clean and healthy, reducing pollution and health risks.

8. Advantage: Dumping of waste is a cost-effective and easy method to dispose of large quantities of waste, especially where land is abundant.

Disadvantage: It can cause environmental pollution, contaminating soil and groundwater, and release harmful gases like methane, contributing to climate change.

9. Recycling of waste is helpful because it reduces the consumption of new raw materials, conserves energy and minimizes environmental pollution. An example of recycling is converting used paper into new paper products.

10. One initiative is the **Swachh Bharat Mission**, which promotes cleanliness, encourages proper waste disposal, and aims to improve waste management across India.

11. Reuse of waste means using materials again instead of discarding them, reducing the need for new resources and minimizing waste.

Example: Old glass bottles can be cleaned and used again to store liquids, or old newspapers can be used for packing or making paper bags.

12. Asbestos: Prolonged exposure can cause lung diseases such as asbestosis, lung cancer, and mesothelioma.

Lead: Inhalation or ingestion can damage the nervous system, cause kidney problems, and lead to developmental issues, especially in children.

13. 1. Environmental Impact: They are non-biodegradable and take hundreds of years to decompose, causing pollution and landfill buildup.

2. Harm to Wildlife: Animals may mistake them for food, leading to ingestion that can harm or kill them.

3. **Chemical Leaching:** Styrofoam can release harmful chemicals into food and drinks, posing health risks to humans.

14. Ragpickers help in **waste management** by collecting discarded materials like metal, glass, rubber, and plastics from landfills. These materials are then recycled into new products, reducing waste and promoting recycling.

15. Segregation of waste involves **separating different types of waste** (like glass, metals, paper, and cloth) before disposal. This helps in:

1. **Safe handling:** Each type of waste can be managed appropriately, reducing environmental and health hazards.
2. **Reuse and recycling:** Segregated materials can be reused or recycled effectively, minimizing landfill waste and pollution.

16. (i) Segregation of waste helps by separating recyclable materials from non-recyclable ones, making waste management efficient and reducing landfill waste.
(ii) Glass jars can be reused as storage containers instead of being thrown away.
(iii) Composting produces nutrient-rich compost that improves soil fertility and reduces the need for chemical fertilizers.

17. The increasing waste is a result of rapid industrialization and urbanization.

(i) New types of waste, like packaging materials, are growing and domestic waste volumes are rising.
(ii) E-waste is increasing rapidly, with no proper handling methods in place.
(iii)These issues highlight the urgent need for waste management to ensure a livable environment for future generations.

18. Benefits of composting are as follows :

1. Provides essential nutrients for plant growth.
2. Reduces harmful effects of excessive alkalinity, acidity, or overuse of chemical fertilizers.
3. Improves soil texture, making it easier to cultivate.

4. Regulates soil temperature and prevents erosion.
5. Helps control weed growth in gardens.

19. (b) Sanitary landfill

A sanitary landfill safely disposes of waste by spreading it in layers, compacting, and covering it with soil, preventing health hazards and environmental pollution.

20. (c) Composting

Composting uses **aerobic microorganisms** to decompose organic waste into nutrient-rich compost, making it an eco-friendly method of waste disposal.

21. (c) Decomposition

Organic household waste breaks down naturally through **decomposition** by microorganisms, forming nutrient-rich compost.

22. (c) Humus

Composting adds **humus** to the soil, improving its fertility, texture, and water-holding capacity.

23. (i) **Plant more trees** to absorb carbon dioxide, release oxygen, improve air quality, provide wildlife habitats, and prevent soil erosion.
(ii) **Use organic manure** to enhance soil health naturally, reduce soil degradation, and minimize water pollution from chemicals.
(iii) **Use mass transport systems** to reduce traffic congestion, lower greenhouse gas emissions, and decrease air pollution, making cities more sustainable.

24. The processes involved in reducing waste are:

1. **Reduce:** Minimize waste by using fewer resources and avoiding unnecessary products.
2. **Reuse:** Extend the life of items by using them multiple times or repurposing them.

3. **Recycle:** Convert waste materials into new products, saving raw materials and energy.
4. **Composting:** Decompose organic waste to create nutrient-rich compost for gardening and agriculture.
5. **Waste Segregation:** Separate waste into biodegradable, recyclable, and hazardous types for proper treatment and disposal.

25. Composting: It is the process of decomposing organic waste, like food scraps, leaves, and plant material, into nutrient-rich compost that can be used as a natural fertilizer.

Advantages:

1. **Improves soil fertility:** Compost enriches the soil with essential nutrients, promoting healthy plant growth and improving soil structure.
2. **Reduces landfill waste:** Composting decreases the amount of organic waste sent to landfills, reducing methane emissions and supporting environmental sustainability.